Lacunary Endpoint Generators: Nonuniqueness at Both Orlicz Boundaries

Candidate counterexample packet for human verification
Run `fa_banach_001`, lane 17, model GPT5.6

11 August 2026

Status: candidate counterexample, likely valid. The packet gives explicit non-quasi-equivalent generators at both endpoint scales. At the quadratic logarithmic endpoint it contradicts Theorem 4.2 of Braverman (1993) as printed and the uniqueness conclusion repeated in Remark 5.1 of Astashkin's 2024 survey. Expert review is required before any public novelty claim.

Abstract

Fix $1 \le p < 2$. For a broad class of increasing functions W, we construct two pairs of symmetric L_p random variables. The two members of each pair have non-quasi-equivalent tails, but their independent copies span the same Orlicz sequence space in L_p. The two Orlicz functions have endpoint asymptotics

$$M_-(t) \asymp \frac{t^p}{W(\log(1/t))}, \qquad M_+(t) \asymp t^2 W(\log(1/t)).$$

In particular this proves nonuniqueness for $t^p/\log^\alpha(e/t)$ and $t^2\log^\alpha(e/t)$ for every $\alpha > 0$. Taking $W(s) = s$ in the upper construction gives a generator of $\ell_{t^2\log(e/t)}$ whose lacunary tail is not comparable with x^{-2}. This is a direct counterexample to the quadratic-boundary implication in the cited 1993 theorem.

1 Source question and the published endpoint claim

The source target is S. V. Astashkin, *On subspaces of Orlicz spaces spanned by independent copies of a mean zero function*, arXiv:2407.14870 [1]. On source PDF page 3 it recalls the following natural question.

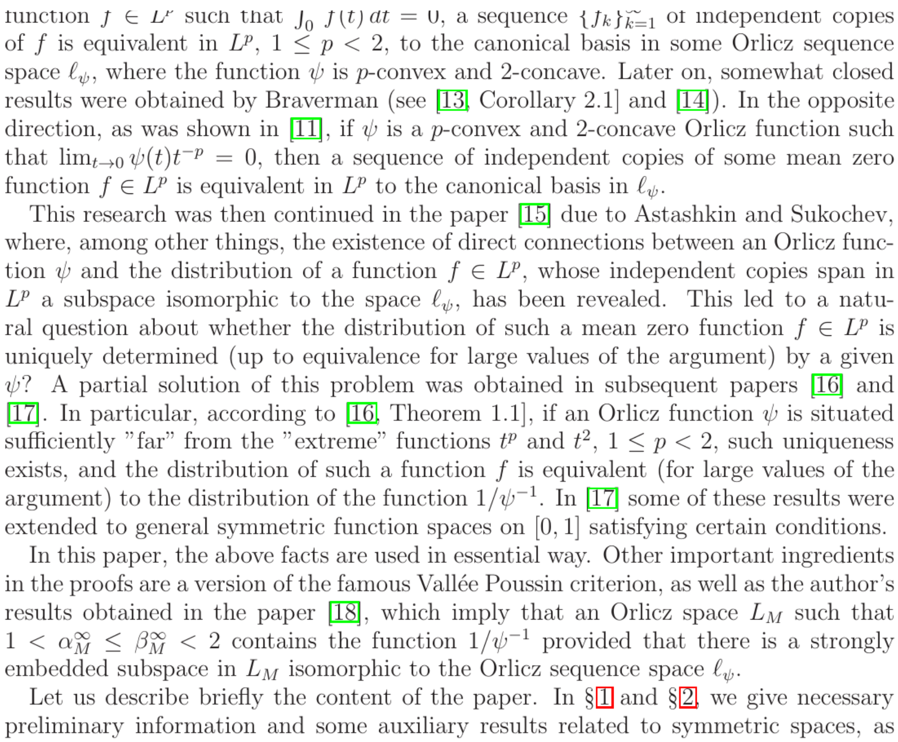
Transcription. If $1 \le p < 2$ and a mean-zero $f \in L_p$ has independent copies spanning the canonical basis of an Orlicz sequence space ℓ_ψ, is the distribution of f uniquely determined, up to equivalence for large values of the argument, by ψ? The source notes that only partial solutions are known.

Astashkin's survey [2, Remark 5.1] states, by reference to Braverman's Theorem 4.2 [4], that the distribution is unique for

$$\psi(t) = t^2 \log_2(2/t).$$

The precise theorem in the cited paper is reproduced in Figure 2. With $h(x)$ slowly varying and

$$\Psi_2(t) = t^2 \int_0^{1/t} \frac{h(y)}{y}\, dy,$$


function $f \in L^p$ such that $\int_0^1 f(t)\,dt = 0$, a sequence $\{f_k\}_{k=1}^\infty$ of independent copies of f is equivalent in L^p, $1 \le p < 2$, to the canonical basis in some Orlicz sequence space ℓ_ψ, where the function ψ is p-convex and 2-concave. Later on, somewhat closed results were obtained by Braverman (see [13, Corollary 2.1] and [14]). In the opposite direction, as was shown in [11], if ψ is a p-convex and 2-concave Orlicz function such that $\lim_{t\to 0} \psi(t)t^{-p} = 0$, then a sequence of independent copies of some mean zero function $f \in L^p$ is equivalent in L^p to the canonical basis in ℓ_ψ.

This research was then continued in the paper [15] due to Astashkin and Sukochev, where, among other things, the existence of direct connections between an Orlicz function ψ and the distribution of a function $f \in L^p$, whose independent copies span in L^p a subspace isomorphic to the space ℓ_ψ, has been revealed. This led to a natural question about whether the distribution of such a mean zero function $f \in L^p$ is uniquely determined (up to equivalence for large values of the argument) by a given ψ? A partial solution of this problem was obtained in subsequent papers [16] and [17]. In particular, according to [16, Theorem 1.1], if an Orlicz function ψ is situated sufficiently "far" from the "extreme" functions t^p and t^2, $1 \le p < 2$, such uniqueness exists, and the distribution of such a function f is equivalent (for large values of the argument) to the distribution of the function $1/\psi^{-1}$. In [17] some of these results were extended to general symmetric function spaces on $[0, 1]$ satisfying certain conditions.

In this paper, the above facts are used in essential way. Other important ingredients in the proofs are a version of the famous Vallée Poussin criterion, as well as the author's results obtained in the paper [18], which imply that an Orlicz space L_M such that $1 < \alpha_M^\infty \le \beta_M^\infty < 2$ contains the function $1/\psi^{-1}$ provided that there is a strongly embedded subspace in L_M isomorphic to the Orlicz sequence space ℓ_ψ.

Let us describe briefly the content of the paper. In §1 and §2, we give necessary preliminary information and some auxiliary results related to symmetric spaces, as

Figure 1: The source paper's uniqueness question and its statement that only partial answers were known, rendered from source PDF page 3.

it claims equivalence between the two-sided coefficient estimate for ℓ_{Ψ_2} and the pointwise tail estimate $\mathbb{P}\{|X| \ge x\} \asymp x^{-2}h(x)$. One may take $h = 1$ for large x and modify it near zero; then $\Psi_2(t) \asymp t^2 \log(e/t)$.

2 Definitions and the disjointification reduction

For a random variable Z, write

$$n_Z(\tau) := \mathbb{P}\{|Z| > \tau\}, \qquad \tau > 0.$$

The robust failure of quasi-equivalence used below is

$$\text{for every } C \ge 1, \qquad \limsup_{\tau\to\infty} \frac{n_X(C\tau)}{n_Y(\tau)} = \infty. \tag{1}$$

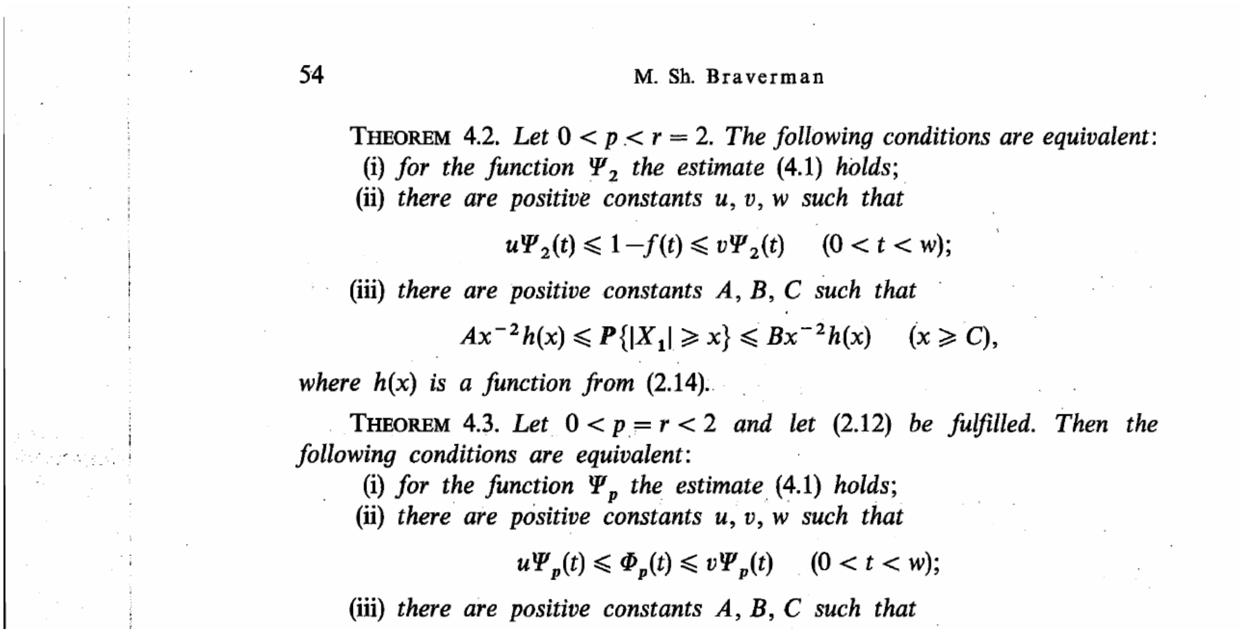
This is exactly the separation condition used in the known lower-endpoint example [2, Theorem 5.3].

Put

$$N_p(u) := \begin{cases} u^2, & 0 \le u \le 1, \\ \dfrac{2}{p}u^p + 1 - \dfrac{2}{p}, & u \ge 1. \end{cases} \tag{2}$$

This is an Orlicz function and $N_p(u) \asymp \min\{u^p, u^2\}$. Thus the function Orlicz space $L_{N_p}(0,\infty)$ is $L_p + L_2$ up to equivalence of norms. For a positive random variable Z, define

$$M_Z(t) := \mathbb{E}N_p(tZ). \tag{3}$$



54 M. Sh. Braverman

THEOREM 4.2. *Let $0 < p < r = 2$. The following conditions are equivalent*:
(i) *for the function Ψ_2 the estimate* (4.1) *holds*;
(ii) *there are positive constants u, v, w such that*

$$u\Psi_2(t) \leqslant 1 - f(t) \leqslant v\Psi_2(t) \qquad (0 < t < w);$$

(iii) *there are positive constants A, B, C such that*

$$Ax^{-2}h(x) \leqslant P\{|X_1| \geqslant x\} \leqslant Bx^{-2}h(x) \qquad (x \geqslant C),$$

where $h(x)$ is a function from (2.14).

THEOREM 4.3. *Let $0 < p = r < 2$ and let* (2.12) *be fulfilled. Then the following conditions are equivalent*:
(i) *for the function Ψ_p the estimate* (4.1) *holds*;
(ii) *there are positive constants u, v, w such that*

$$u\Psi_p(t) \leqslant \Phi_p(t) \leqslant v\Psi_p(t) \qquad (0 < t < w);$$

(iii) *there are positive constants A, B, C such that*

Figure 2: Braverman's Theorem 4.2, journal page 54 (physical PDF page 10).

Disjoint copies of Z in $L_{N_p}(0,\infty)$ span ℓ_{M_Z} exactly in Luxemburg modulars. The Johnson–Schechtman/Rosenthal disjointification estimate [5], in the form recorded in the source literature [1, § 2.4], implies that if $f = \varepsilon Z$, where ε is an independent Rademacher sign, then independent copies of f in $L_p(0,1)$ are equivalent to the canonical basis of ℓ_{M_Z}. Consequently it is enough to construct positive $X, Y \in L_p$ such that $M_X \asymp M_Y$ near zero and (1) holds.

Notice also that M_Z is p-convex and 2-concave: after the substitutions $t = u^{1/p}$ and $t = u^{1/2}$, these properties hold termwise for $N_p(tZ)$ and are preserved by integration.

3 Proof intuition

The modular (3) smooths a distribution across the threshold $Z = 1/t$. At the lower endpoint, its main term is the p-moment of the tail; at the upper endpoint, it is the truncated second moment below the threshold. Pointwise tail data can therefore be moved between extremely distant atoms without changing the smoothed modular by more than a constant.

We put atoms at heights $A_k = \exp(2^k)$. One distribution uses every height, whereas the other uses only odd-indexed heights. Since adjacent logarithmic scales differ only by a factor two (or four on the odd grid), both modulars remain comparable. In the enormous gap between A_{2j+1} and A_{2j+2}, however, the full distribution still sees its A_{2j+2} atom while the odd distribution must wait until A_{2j+3}. The resulting tail ratio grows doubly exponentially and defeats every fixed rescaling.

4 The geometrically doubling construction

Theorem 1 (Two endpoint nonuniqueness families). *Let $1 \le p < 2$. Suppose $W : [2,\infty) \to (0,\infty)$ is nondecreasing and there are constants $1 < \lambda \le \Lambda < \infty$ such that*

$$\lambda \le \frac{W(2s)}{W(s)} \le \Lambda, \qquad s \ge 2. \tag{4}$$

Then there are symmetric mean-zero random variables $f_-, g_-, f_+, g_+ \in L_p(0,1)$ with the following properties.

(i) *The independent copies of f_- and g_- are both equivalent in L_p to the canonical basis of the same Orlicz sequence space ℓ_{M_-}, where*

$$M_-(t) \asymp \frac{t^p}{W(\log(1/t))} \qquad (t \downarrow 0).$$

(ii) *The independent copies of f_+ and g_+ are both equivalent in L_p to the canonical basis of the same Orlicz sequence space ℓ_{M_+}, where*

$$M_+(t) \asymp t^2 W(\log(1/t)) \qquad (t \downarrow 0).$$

(iii) *Each pair has non-quasi-equivalent distributions in the strong sense of* (1).

The same conclusion holds when (4) *is imposed only for all sufficiently large s.*

Proof. Set

$$u_k := 2^k, \qquad A_k := e^{u_k}, \qquad w_k := W(u_k), \qquad k \geq 1.$$

Choose positive normalizing constants so that the following masses sum to at most one, and put all unused probability at zero. Define positive random variables X_-, Y_-, X_+, Y_+ by

$$\mathbb{P}\{X_- = A_k\} = c_- w_k^{-1} A_k^{-p}, \qquad k \geq 1, \qquad \mathbb{P}\{Y_- = A_k\} = d_- w_k^{-1} A_k^{-p}, \quad k \text{ odd}, \tag{5}$$

$$\mathbb{P}\{X_+ = A_k\} = c_+ w_k A_k^{-2}, \qquad k \geq 1, \qquad \mathbb{P}\{Y_+ = A_k\} = d_+ w_k A_k^{-2}, \quad k \text{ odd}. \tag{6}$$

Condition (4) gives $w_{k+1}/w_k \in [\lambda, \Lambda]$. Therefore

$$\mathbb{E}X_-^p = c_- \sum_{k\geq 1} w_k^{-1} < \infty, \qquad \mathbb{E}X_+^p = c_+ \sum_{k\geq 1} w_k e^{-(2-p)u_k} < \infty,$$

and the same holds for the odd-grid variables. Thus all four are in L_p.

We first estimate the full-grid modulars. Let $s = \log(1/t)$ and choose m so that $u_m \leq s < u_{m+1}$. Since $N_p(v) \asymp \min(v^p, v^2)$,

$$M_{X_-}(t) \asymp t^2 \sum_{k\leq m} w_k^{-1} e^{(2-p)u_k} + t^p \sum_{k>m} w_k^{-1}, \tag{7}$$

$$M_{X_+}(t) \asymp t^2 \sum_{k\leq m} w_k + t^p \sum_{k>m} w_k e^{-(2-p)u_k}. \tag{8}$$

For the lower modular, the second sum is a geometric tail and

$$\sum_{k>m} w_k^{-1} \asymp w_{m+1}^{-1} \asymp W(s)^{-1}. \tag{9}$$

The summands $w_k^{-1} e^{(2-p)u_k}$ in the first sum are eventually increasing by a factor at least two, because $u_{k+1} - u_k = u_k$ and the exponential dominates the fixed factor w_{k+1}/w_k. Finitely many initial indices can be absorbed into the constant, so

$$\sum_{k\leq m} w_k^{-1} e^{(2-p)u_k} \lesssim w_m^{-1} e^{(2-p)u_m}.$$

Using $W(s)/w_m \le \Lambda$ and $s \ge u_m$, the ratio of the resulting first term in (7) to $t^p/W(s)$ is at most

$$\frac{W(s)}{w_m} e^{-(2-p)(s-u_m)} \le \Lambda.$$

Together with (9), this proves

$$M_{X_-}(t) \asymp \frac{t^p}{W(s)}. \tag{10}$$

For the upper modular, geometric growth gives

$$\sum_{k \le m} w_k \asymp w_m \asymp W(s), \tag{11}$$

which already supplies both bounds at the target scale. The terms $w_k e^{-(2-p)u_k}$ are eventually decreasing by a factor at least two, so the tail in (8) is dominated by its first term. Since $w_{m+1}/W(s) \le \Lambda$ and $s < u_{m+1}$,

$$\frac{t^p w_{m+1} e^{-(2-p)u_{m+1}}}{t^2 W(s)} = \frac{w_{m+1}}{W(s)} e^{-(2-p)(u_{m+1}-s)} \le \Lambda.$$

Consequently

$$M_{X_+}(t) \asymp t^2 W(s). \tag{12}$$

The identical proof applies on the odd grid. Indeed its logarithmic scales are $u_{2j+1} = 2 \cdot 4^j$, consecutive scales have ratio four, and

$$\lambda^2 \le \frac{W(4s)}{W(s)} \le \Lambda^2.$$

For s between two consecutive odd scales, monotonicity and the last display make $W(s)$ comparable with the preceding odd-grid weight. Hence

$$M_{Y_-}(t) \asymp M_{X_-}(t), \qquad M_{Y_+}(t) \asymp M_{X_+}(t). \tag{13}$$

It remains to prove robust tail separation. Fix $C \ge 1$. For all large j, the gap A_{2j+2}/A_{2j+1} exceeds $2C$; choose

$$\tau_j := \frac{A_{2j+2}}{2C}.$$

Then both τ_j and $C\tau_j$ lie strictly between A_{2j+1} and A_{2j+2}. The full-grid tail at $C\tau_j$ contains its A_{2j+2} atom, whereas the first atom in the odd-grid tail at τ_j is A_{2j+3}. The rest of the odd tail is at most a fixed multiple of that first atom, again because the double exponential dominates the geometric variation of w_k. For the lower pair the ratio of the decisive masses is bounded below by a fixed positive multiple of

$$\frac{w_{2j+3}}{w_{2j+2}} e^{p(u_{2j+3}-u_{2j+2})} \ge \lambda e^{p u_{2j+2}} \longrightarrow \infty,$$

and for the upper pair it is bounded below by a fixed positive multiple of

$$\frac{w_{2j+2}}{w_{2j+3}} e^{2(u_{2j+3}-u_{2j+2})} \ge \Lambda^{-1} e^{2u_{2j+2}} \longrightarrow \infty.$$

This proves (1) for both pairs.

Finally take independent Rademacher signs and set $f_- = \varepsilon X_-$, $g_- = \varepsilon Y_-$, $f_+ = \varepsilon X_+$, and $g_+ = \varepsilon Y_+$. They are symmetric, mean zero, and in L_p. The disjointification reduction preceding the theorem, together with (13), proves the asserted equivalence of their independent-copy sequences. If (4) holds only eventually, discard finitely many initial atoms; this changes none of the asymptotic conclusions. □

Corollary 2 (Logarithmic and log–log endpoints)**.** *For every $1 \le p < 2$, $\alpha > 0$, and $\beta \in \mathbb{R}$, there are two non-quasi-equivalent generators for each of the Orlicz sequence spaces with small-argument asymptotics*

$$\frac{t^p}{\log^\alpha(e/t)\,[\log\log(e^e/t)]^\beta} \quad \textit{and} \quad t^2\log^\alpha(e/t)\,[\log\log(e^e/t)]^\beta.$$

In particular one may take $\beta = 0$.

Proof. After an irrelevant modification on a bounded interval, take $W(s) = s^\alpha[\log(e+s)]^\beta$. It is eventually nondecreasing and $W(2s)/W(s) \to 2^\alpha > 1$, so Theorem 1 applies. □

5 Explicit contradiction to the quadratic endpoint claim

Take $W(s) = s$ and either member of the upper pair. Then

$$M_Z(t) = \mathbb{E}N_p(tZ) \asymp t^2\log(e/t). \tag{14}$$

By disjointification, the independent copies of the symmetric mean-zero version of Z satisfy Braverman's coefficient estimate (4.1) for $\Psi_2(t) \asymp t^2\log(e/t)$.

Yet the tail is not comparable with x^{-2}. For example, just above one atom choose $x_j = 2A_j = 2A_{j+1}^{1/2}$; for large j this still lies below A_{j+1}. The tail is dominated by the A_{j+1} atom, so

$$n_Z(x_j) \lesssim u_{j+1}A_{j+1}^{-2}, \qquad x_j^{-2} = \tfrac{1}{4}A_{j+1}^{-1},$$

and hence $n_Z(x_j)/x_j^{-2} \to 0$. Thus condition (i) of the printed Theorem 4.2 holds while condition (iii), with $h = 1$ at infinity, fails. The full/odd pair gives the stronger failure of uniqueness (1) asserted in Astashkin's survey.

The mechanism is the familiar Tauberian boundary obstruction: at regular variation index two, the characteristic-function or coefficient estimate controls a truncated second moment, but without an anti-lacunarity assumption it does not recover a pointwise tail. The proof above does not rely on this interpretation; it verifies the two conflicting conditions directly.

6 Verification, novelty search, and limitations

The proof is analytic. The included script `code/check_endpoint_examples.py` evaluates all modular sums in logarithmic arithmetic. It tests $p \in \{1, 1.3, 1.8\}$, three choices of $W(s) = s^\alpha\log^\beta(e+s)$, both endpoints, both grids, and 37 threshold scales per case. All 1332 modular ratios stay finite and bounded on the tested range. It also prints tail-separation lower bounds growing from more than 10^{36} to more than 10^{14000}. The command is

```
conda run --no-capture-output -n sandbox python \
  code/check_endpoint_examples.py
```

The computation is a stress test only; the estimates (7)–(13) are the proof.

A bounded novelty audit was performed through 11 August 2026. The run's registry and solution indexes were searched for arXiv:2407.14870, arXiv:1406.4950, Braverman's theorem, both logarithmic endpoint formulae, "quasi-equivalent distributions," and lacunary/staircase generators. The arXiv/full-source corpus and external primary-source searches used the exact 1993 title and theorem, the 2015 paper [3], and the 2024 survey [2]. No correction or matching upper-endpoint counterexample was found; the 2024 survey still states uniqueness for $t^2\log_2(2/t)$.

The registry does contain a separate candidate packet for arXiv:1406.4950 based on a general Hardy staircase perturbation. That packet explicitly proves only failure of pointwise equivalence at the same quantile and does not claim the rescaling-robust quasi-equivalence separation. The full/odd tail gap in this packet proves the stronger condition (1), so the present result is not a duplicate.

This packet does not characterize all Orlicz functions with unique distribution, and it does not settle the full necessity conjecture under the published quasi-equivalence notion. It gives two broad endpoint families and an explicit counterexample to a claimed endpoint theorem. Novelty confidence is moderate until an expert checks the 1993 theorem's conventions and searches non-arXiv probability literature.

Human-review recommendation. First verify the threshold split in (7) and (8); then check the full/odd tail comparison for an arbitrary fixed C. Finally compare (14) with Braverman's definitions (2.14), (4.1), and Theorem 4.2. These three checks isolate the entire counterexample.

References

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[3] S. Astashkin, F. Sukochev, and D. Zanin, *On uniqueness of distribution of a random variable whose independent copies span a subspace in L_p*, Studia Math. 230:1 (2015), 41–57; arXiv:1406.4950.

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